

Lab 2 - Linguistics Data, Stat 215A, Fall 2026

Introduction

John Nerbonne and William Kretzschmar describe dialectometry as “the measurement of dialect differences, i.e. linguistic differences whose distribution is determined primarily by geography” (Nerbonne & Kretzschmar, 2003). Linguistic differences across geographic regions have been studied in many ways, and both Nerbonne & Kretzschmar 2003 and Nerbonne & Kretzschmar 2006 outline attempts to statistically describe the correlation between dialect and geography. They further delve into a few methods that identify other ways to explain linguistic variations that have ultimately pushed the field forward. In this report, we will primarily focus on the connection between dialects and geography, using the idea of dialectometry to analyze results from the 2003 Harvard Dialect Survey (<http://dialect.redlog.net/>). From the survey results, we would like to know if there is a real difference in the way that individuals say/pronounce/communicate a certain word or idea between geographical contexts in the United States. We wish to know if there are distinct geographic groups who pronounce certain words differently, if these groups can be used to predict how words are said, and if these groups relate to surrounding geography. Through the exploration of geography and dialect groupings, we will be analyzing how well different statistical techniques are performing. Considering the size of the data, we will continue with an assessment of dimension reduction techniques, PCA consideration, and two different clustering methods which will be further evaluated by using data perturbations. Lastly, we will consider the three realms of data science—data, algorithms and analysis, and future data—and will analyze our results through the lens of each.

The Data

In this report we analyze the 2003 Harvard Dialect Survey. This survey asks a series of 121 questions. Each question explores how a participant would pronounce a word, say a phrase, or refer to an item/idea. The survey had a total of 30,934 respondents of all ages, ranging from under 13 years old to over 70 years old. There is at least one respondent from every US state. For this report, we will specifically be looking at lexical differences as opposed to phonetic differences, and therefore, only look at 50-121. No extra step is needed for this, since the data set gives has already dropped questions 1-50.

For most of the lexical questions, the questions usually began with “what do you call...” or “Would you say...” and then it would end with a description or way to say something. Each question usually had many responses labeled a, b, c, etc. In our dataset, a is coded as 1, b is coded as 2, etc. The location of each respondent was also taken. Then, for every question, each answer was tabulated and a map of where each answer was selected can be seen on the website <http://dialect.redlog.net/>.

This data will be extremely helpful in determining for linking dialects to geography. By mapping the data, we will be able to visually see where these words/sayings are used. Then, we can see if our statistical tools will be helpful in determining a true link between dialect and geography, which tools work, which to avoid, and the limitations of each.

Data Cleaning

To clean the data, we standardized geographic labels, resolved location mismatches with spatial checks, imputed missing coordinates, and removed unresolvable or empty responses. Of the original 47,471 respondents, we dropped 1,453 (~3.06%), leaving 46,018.

We first mapped state abbreviations to full names and normalized city names. We removed rows with state “XX”, non-US cities, and 91 rows with unrecognized states and no city. We filled in missing cities for 404 rows using ZIP codes that matched known cities in the same state, and dropped 88 rows that could not be filled.

We then checked that each respondent fell inside their reported state. Because the simplified polygons place NYC boroughs inside New Jersey, we assigned those respondents to NY. For the 387 rows located in a different state, we trusted the self-reported city and state over the ZIP-based coordinates (which were entered by GSIs), correcting the state or moving the point to its city; 59 rows could not be resolved and were removed. Points outside every state were moved to their city’s median location, or snapped to the nearest edge of their state for coastal cities, and 6 rows too far away were dropped. Finally, we filled coordinates for 160 rows from matching ZIP codes, dropped 94 rows we could not locate, and removed 1,032 respondents who answered no questions. Michigan’s polygon remains imperfect along its shoreline, which we would fix with more time.

	Metric	Count	Percentage
0	Dropped (No Answers)	1,032	2.17%
1	Total Dropped	1,453	3.06%
2	Final Sample Size	46,018	96.94%

Table 1: Cleaned Data Details

Exploratory Data Analysis

Answers to Question 65 by response: What do you call the bug that glows at night?

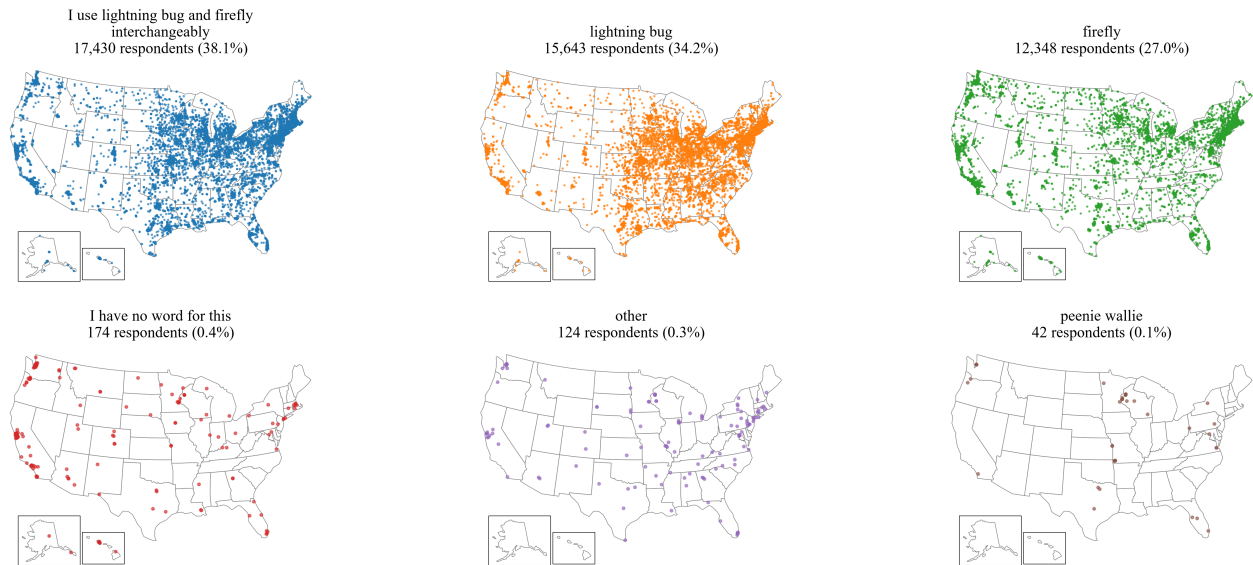


Figure 1: Map of All Responses to Question 65

Answers to Question 66 by response: What do you call the miniature lobster found in lakes and streams?

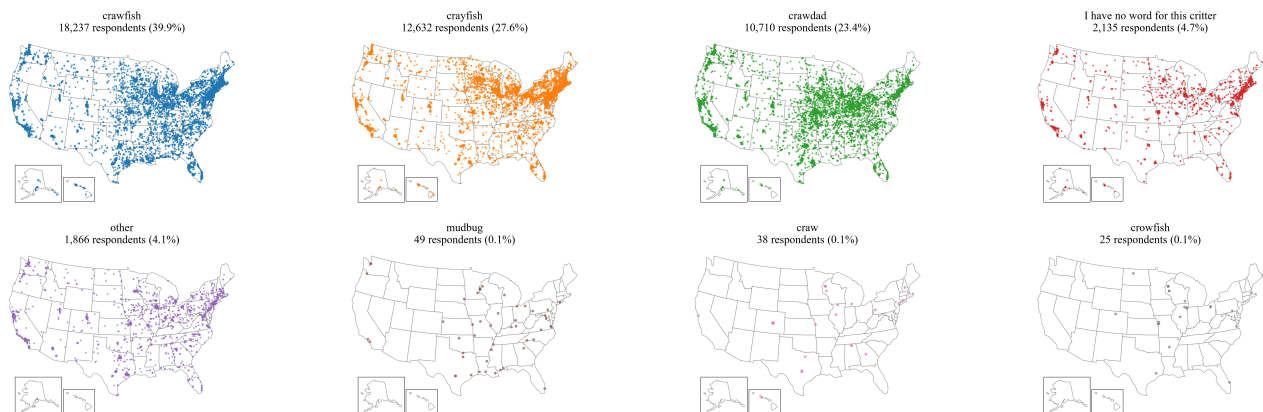


Figure 2: Map of Individual Responses to Question 65

Answers to Question 65: What do you call the bug that glows at night?

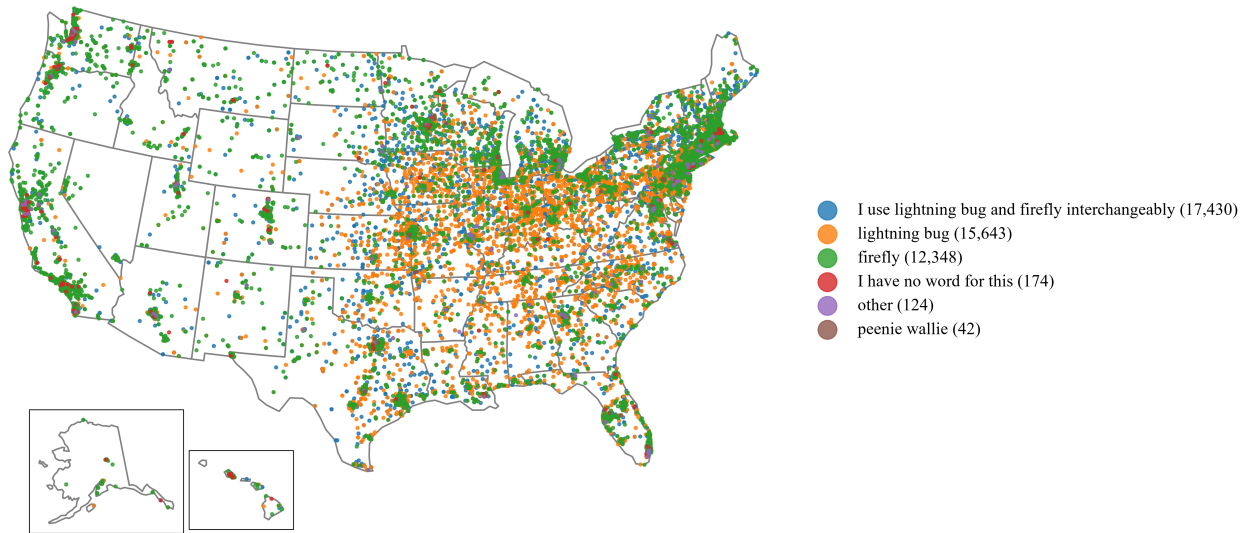


Figure 3: Map of All Responses to Question 66

Answers to Question 66: What do you call the miniature lobster found in lakes and streams?

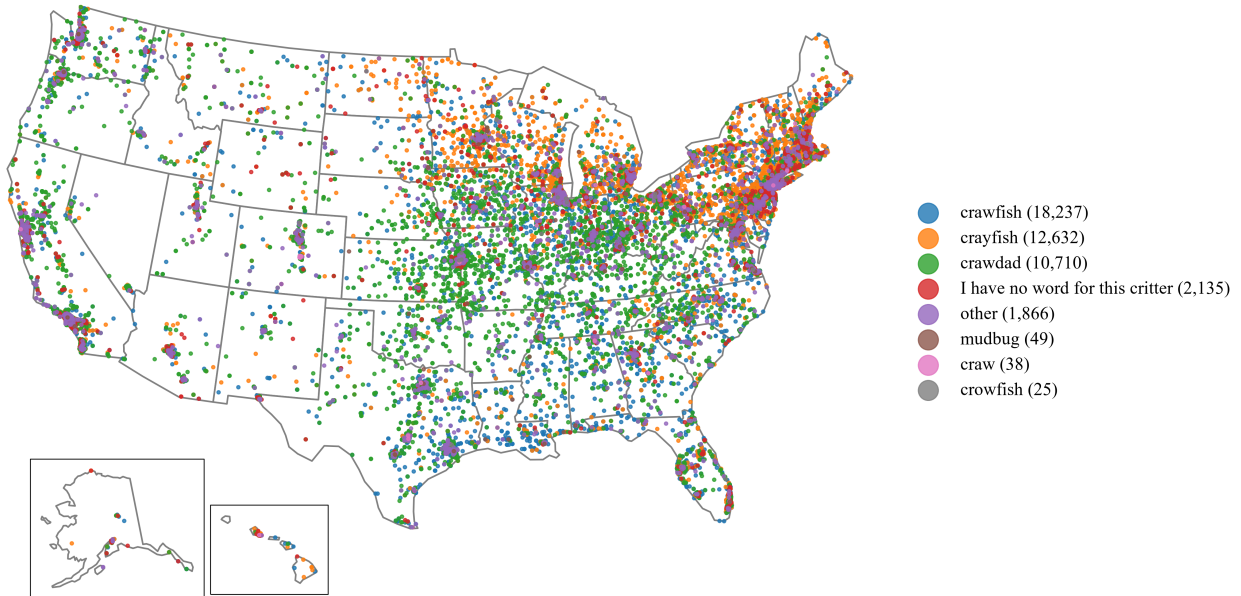


Figure 4: Map of All Responses to Question 65

When looking at the two graphs that map the answers to questions 65 and 66, we see that in both maps, distinct geographical groups begin to form. Broadly, considering just those who answered the question, we can see that many responses are clustered along the north eastern part of the US. Then, we see another cluster of responses along the west coast. Then, we also see clusters around capitol cities such as Pheonix AZ, Denver CO, Salt Lake City UT, and Albuquerque NM. This trend aligns with any searched population density map of the US. So, upon first glance it seems that responses are quite widespread across the US and align well with the expected US population.

The next trend that we see are certain colors having higher frequencies in certain areas. In the glowing bug question, we see the highest concentration of the term “firefly” (27%) in cities and along the top of north eastern states. We also see it heavily used on the west coast. However, we see the use of the term “lightning bug” (34.2%) spread more densely throughout the eastern states, even in places that may be a bit more rural. Interestingly, we also see the same ubiquitous trend for those who “use lightning bug and firefly interchangeably” (38.1%). This may be due to the dialects being more distinct at one point in history, but then the spreading and sharing of terms led to the conglomeration of dialectal words.

In the mini lobster question, we see that “crawfish” is the most common response (39.9%), and it seems to follow a similar trend with the “lightning bug” category from the previous question. The other two most common responses were “crayfish” (27.6%) and “crawdad” (23.4%). The responses indicating “crayfish” seem to have a trend more toward the north eastern states in the “rust belt” and New England areas. The response indicating “crawdad” seems to trend a bit more south, reaching through the entire midwest and into the south east.

We also see that there are some responses that give rare answer such as “crowfish” for the mini lobster question (with only 0.1% of responses) and “peenie wallies” for the glowing bug question (with only 0.1% of responses).

Q065_grouped Q066_grouped	both	firefly	lightning bug	other
I have no word for this critter	33.0	34.0	30.2	2.8
crawdad	31.8	22.9	44.6	0.7
crawfish	40.2	24.9	34.5	0.4
crayfish	39.4	33.8	26.3	0.6
other	49.5	17.7	30.3	2.5

Table 2: Associations Between Question 65 and 66 Responses

	LR statistic	df	p-value	Pseudo R ² (state + Q66)	Pseudo R ² gained from Q66
Q66 → Q65, controlling for state	1072.811982	12	< 0.0001	0.100724	0.010449

Table 3: Multinomial Logit Predicting Question 65 from 66

A correlation table shows that there seem to be preferences in terms of what words someone chooses over others. This does not account for geography, but it does show that some responses may “go together.” For example, if someone in this dataset chose “crawdad” for question 66, they are almost twice as likely to choose “lightning bug” (44.6%) over “firefly” (22.9%). However, although this is not a predictive algorithm and does not prove causation. In speech, there are so many confounding variables such as geography, age, and parent’s background. It does not distinguish between a true dialectic link or simply a geographic one. However, it does seem that there is evidence to support that one may help predict the other in some ways.

Next, we tried to see how a multinomial logit model may observe the relationship between these variables while controlling a bit for spatial confounding to see if there is a predictive dialectic link between question 65 and question 66 that is unattached to geography. This revealed a very small p-value that indicates question 66 is a statistically significant predictor of question 65, but, as can be seen by the Pseudo R² value, it only accounts for about 1% of your predictive accuracy of the response to question 65. If you knew the state, the predictive accuracy increases almost ten-fold.

This exploration means that in order to try to predict responses, geography is going to play a key role.

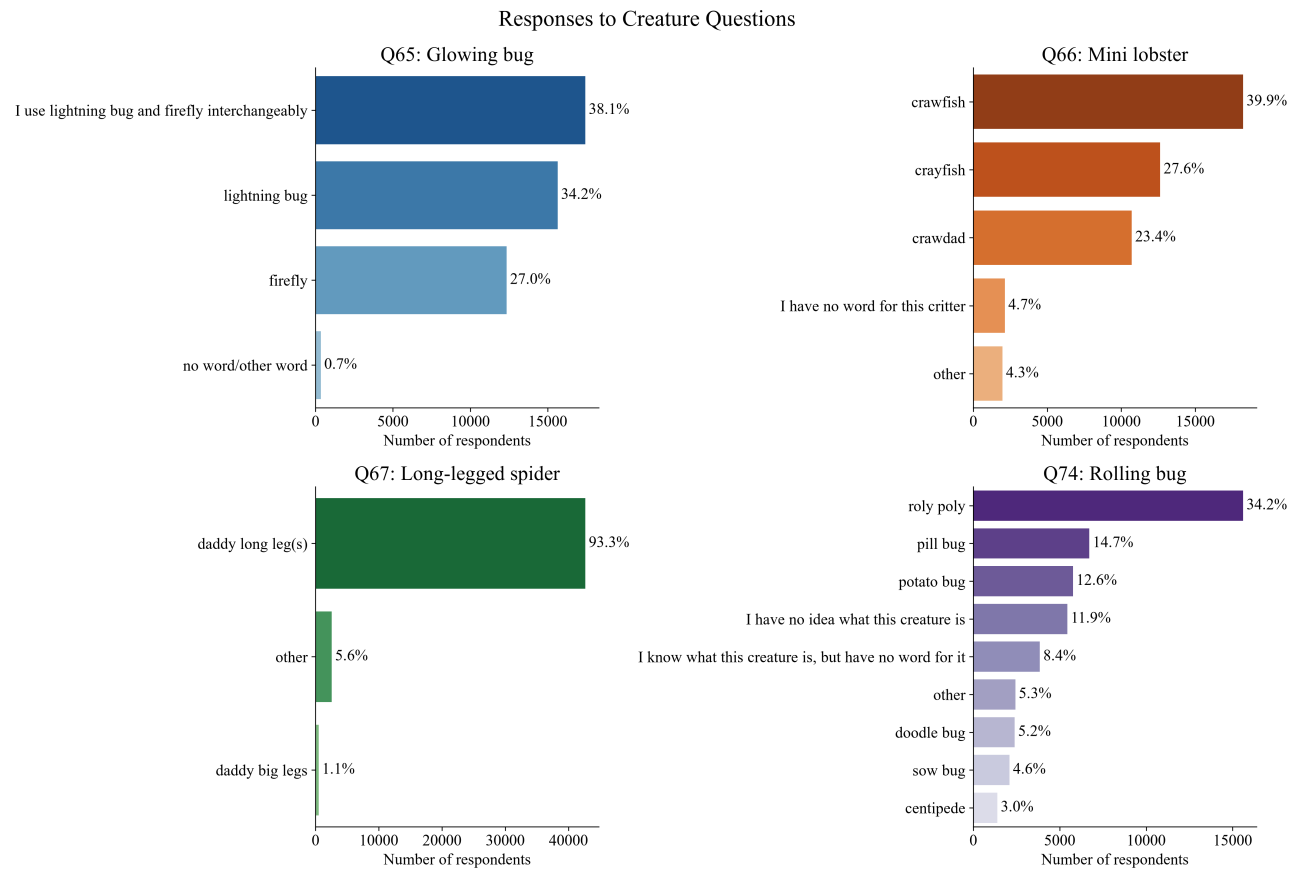
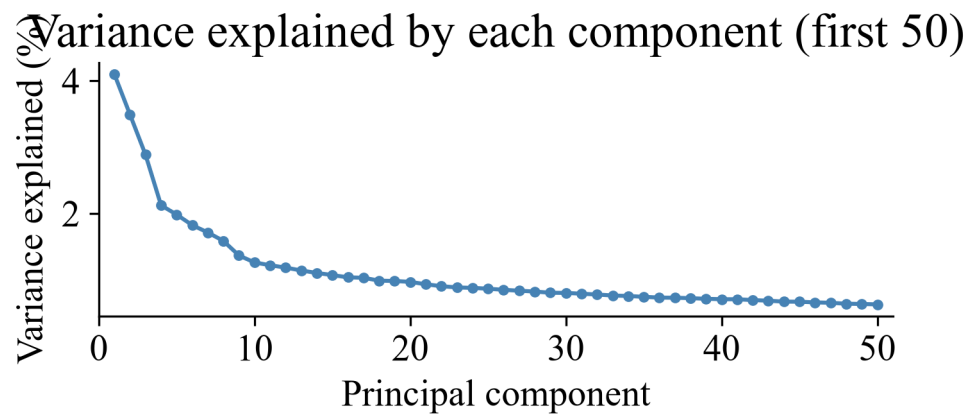


Figure 5: Creature Question Observations

When looking at more questions, it is interesting to note that some responses, such as for the long-legged spider, do not have much variation. That is, there is a clear word that most people use to describe this creature. In other cases, there are a few words that stand out about equally and a smaller portion of the population does not have or has another word for the creature that was not available on the survey. Interestingly, in the case of the rolling bug, multiple respondents had different words for the bug. Also curious is that many respondents did not know what this creature was. This could be due to the way the question was written or possibly geographic natural habitat range of this animal means that some individuals are less likely to encounter them. This is important to note for the rest of the rest of the report, since we can note that some groups will have little variance and contribute less to any PCA analysis. Further, some responses such as “I have no idea what this creature is,” as seen in question 74, could cause problems, especially since lack of knowledge is not something that we are trying to analyze. If two people from very separate regions claim they do not know what they call a certain creature, then clustering models may draw a false geographic equivalency between them. Lastly, there is variation in the number of dimensions for each question. Question 74 about the rolling bugs has multiple responses that increases the dimensionality and can potentially impact a future model. Questions like this will have to be noted when reducing dimensions.

Dimension Reduction

PCA



42 components explain at least 50% of the variance
78 components explain at least 70% of the variance
107 components explain at least 80% of the variance
156 components explain at least 90% of the variance

Figure 6: PCA Scree Plot

Respondents Colored by PC1 Score (Northeast Dialect)

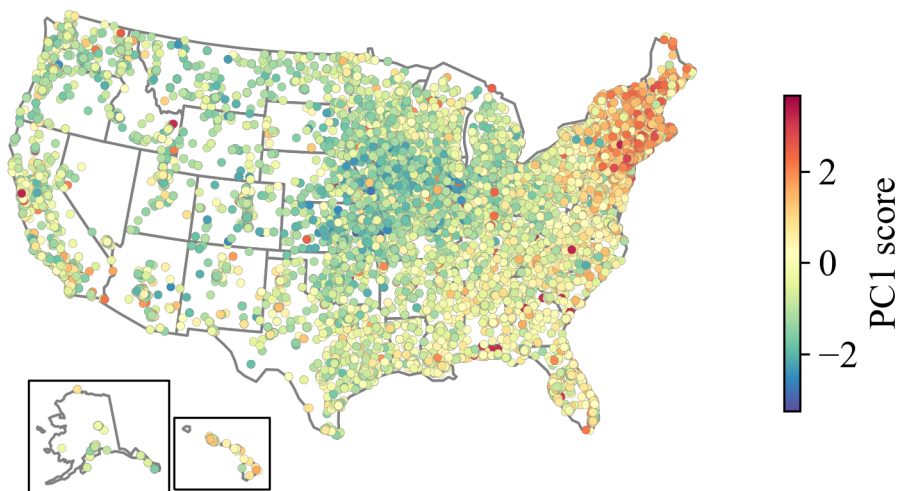


Figure 7: PCA Map of Responses

t-SNE

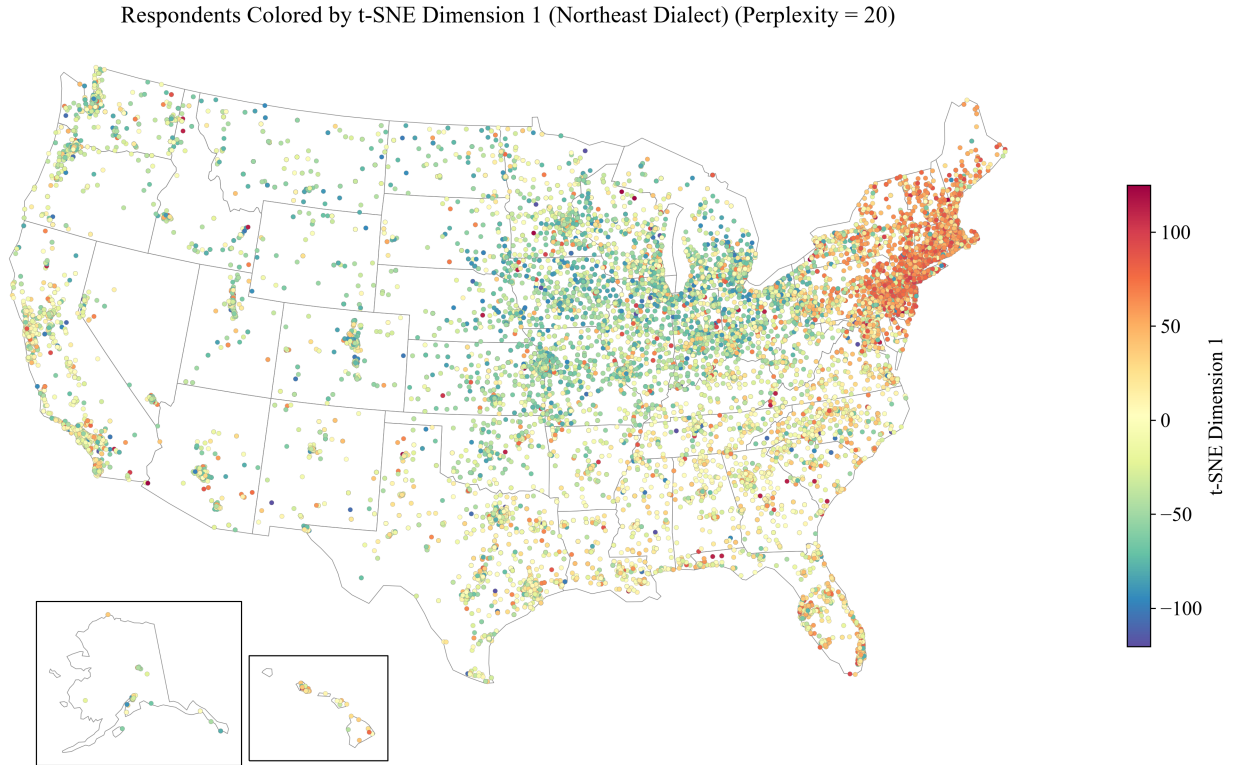


Figure 8: t-SNE Map of Responses

In this section, we did center the data so that we could detect any differences in language from the “average” choices. This way, local trends could seem more distinctive. We decided not to scale so that every answer could have equal importance. Standardizing binary variables inversely weights features by frequency, which can amplify rare choices (“Other” or “Don’t know”) into huge spikes. There was no need to weight any of the answers more heavily.

The original lingData dataset had response codes, but these are quite arbitrary and ordinal. These codes had 1 for ‘a’, 2 for ‘b’, 3 for ‘c’, etc. Running PCA directly on raw integers falsely assumes an ordinal metric space where choice 5 is mathematically “greater than” choice 3 or 4. However, that is not the case, and they are all have the same “value” in context. We used one-hot encoding to ensure that each answer was given its own dimensional space, allowing us to look across all answers for trends in shared responses.

Linear PCA cleanly isolates the primary national variance on PC1, pointing directly to a distinct Northeastern/Mid-Atlantic dialect. For this analysis, we determined the optimal number of dimensions (K) for both the PCA and t-SNE projections. We used a scree plot to see the variance per principle components. There was a clear elbow and flattening at K=10. Therefore, we know that most of the variance is captured with 10 principle components.

However, to more clearly define language areas, and better evaluate non-linear language clusters, we also projected the data using t-SNE on the top 10 PCs. Both PCA and t-SNE produce similar trends, but the t-SNE shows more distinct clusters. Since the t-SNE has a greater focus on local attributes, we can better identify differences in dialect choices. For our graph, we used a perplexity score of 20, which captures a bit more detail.

It as opposed to PCS, the t-SNE more clearly separates the north eastern states, Rust Belt states, and even gives some distinction to the the south east, confirming that there are real regional patterns.

Clustering

K-Means Clustering

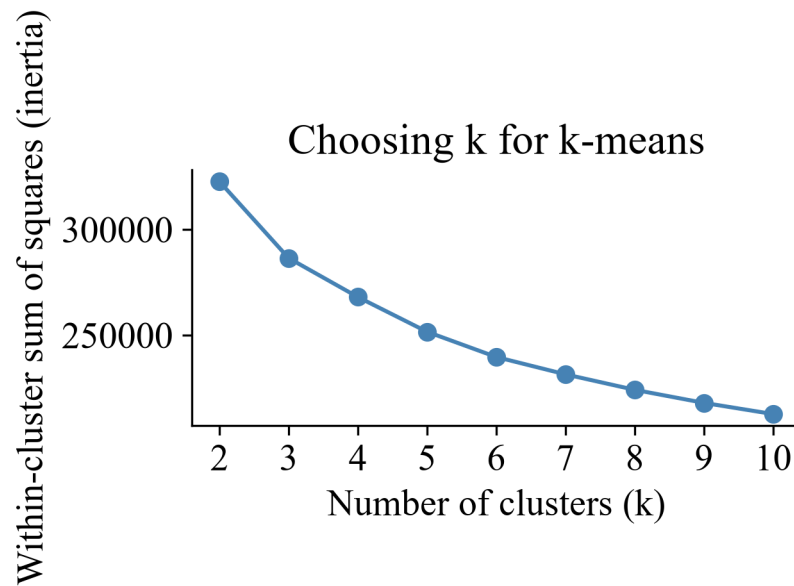


Table 1

k	Silhouette Score
3	0.142
4	0.126
5	0.141
6	0.134
7	0.127
8	0.112

Figure 9: Scree Plot for K-means Table 4: Scree Plot for K-means

K-Means Clusters (k = 3) of Respondents' Answers

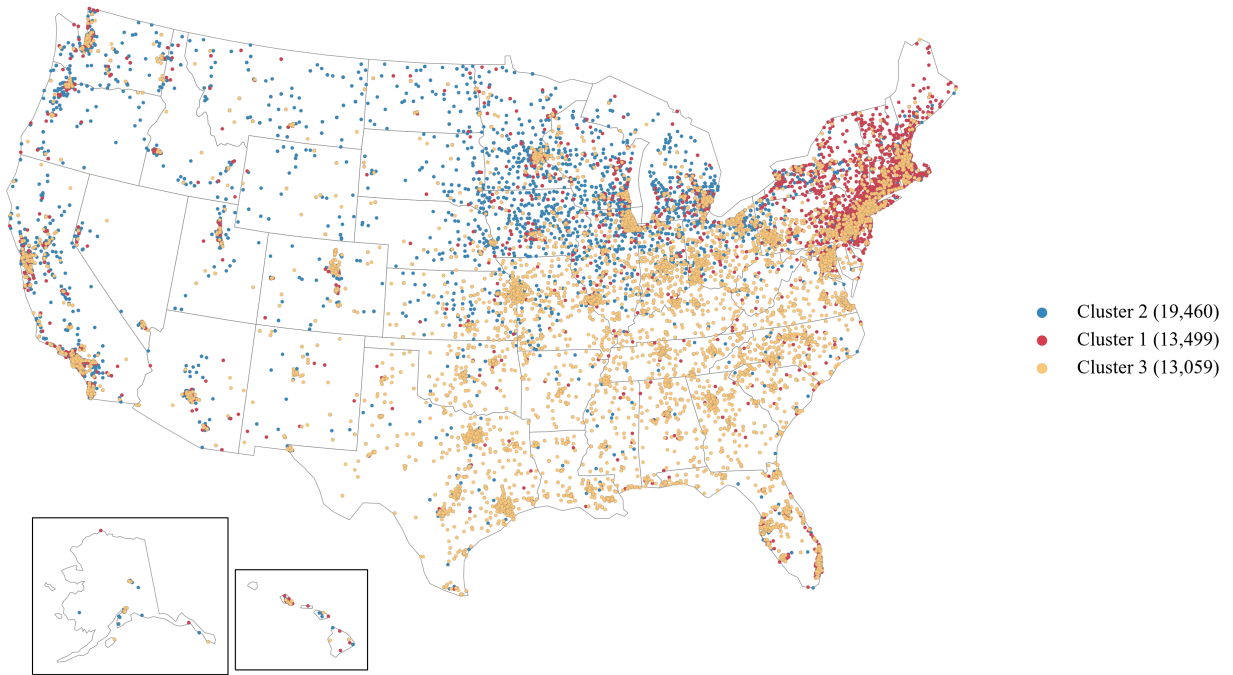


Figure 10: K-Means Cluster Map

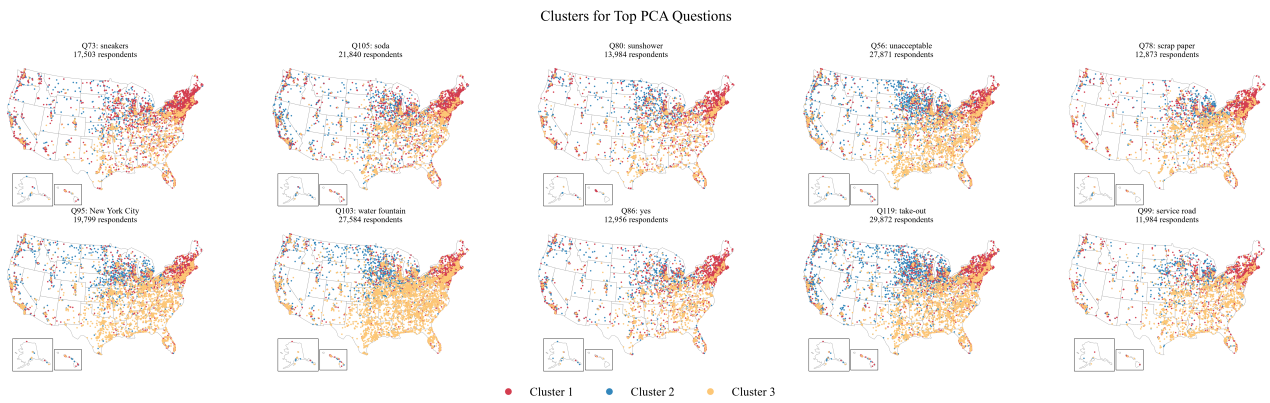


Figure 11: K-Means Cluster Map, Top 10 PCA Questions

Heirarchical Clustering

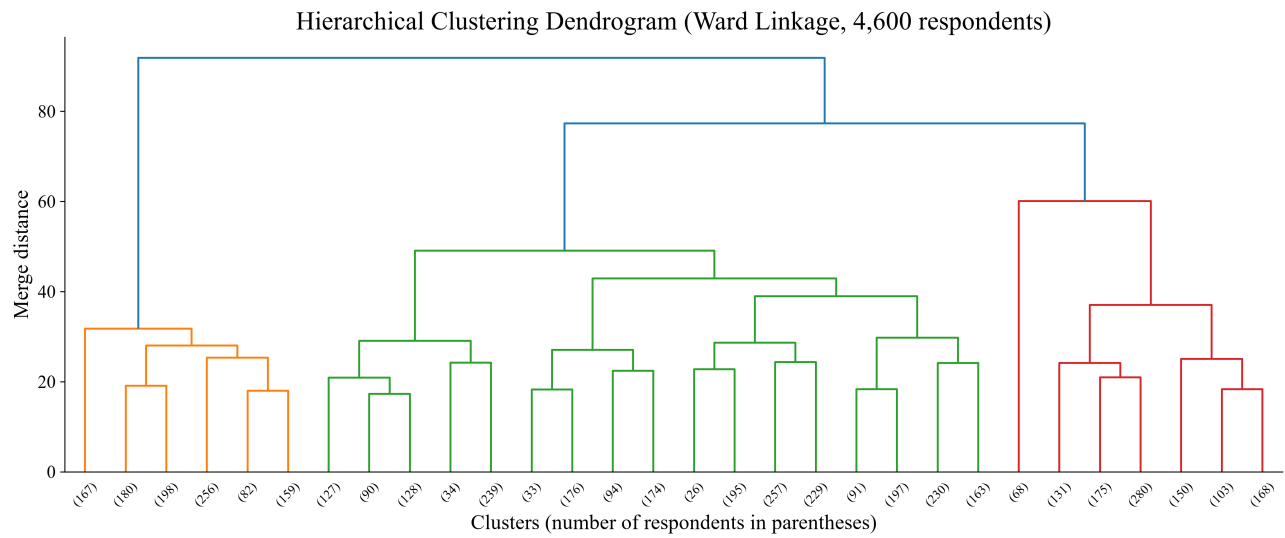


Figure 12: Heirarchical Dendrogram

Hierarchical clusters (Ward, $k = 3$) of 4,600 respondents

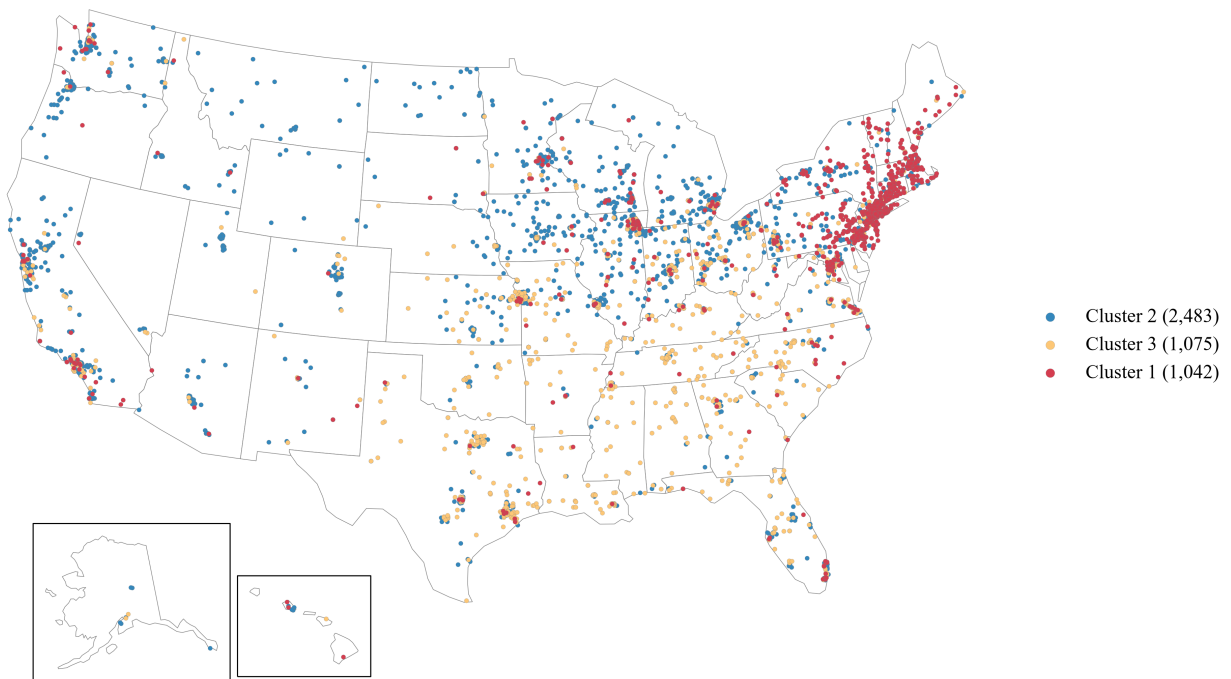


Figure 13: Heirarchical Clusters

Adjusted Rand index (k-means vs. hierarchical): 0.502

k-means	1	2	3
Hierarchical			
1	95.7	1.4	2.9
2	11.3	74.1	14.6
3	7.6	9.7	82.7

Table 5: K-Means and Hierarchical Correlations

We selected $k = 3$ to produce interpretable and stable regional boundaries. To select this value, we first explored which k to use with the elbow method. However, the elbow graph did not have a distinctive curve, so we then looked at the silhouette scores for a few k values. We took the k value with the highest silhouette score, $k = 3$. This cleanly isolated the Northeast corridor, the Southeast, and a broad General American baseline, which directly matches the patterns identified in our PCA and t-SNE projections.

In the k-means clustering graph, we found that there were not completely separate clusters. There was generally a continuum, with distinct regional groups. Urban centers seemed to have tight core clusters, while rural areas and transitional zones blended across boundaries. When looking at individual components, we can see that none of even our top principle components form completely distinct geographic groups from each other. There is always a little bit of overlap. While even the best principal components contribute to the continuum, we would imagine that the smallest principle components would contribute most to it. Questions about the term for gym shoes (question 73) and name of a carbonated beverage (question 105) were the two most likely to make distinctive clusters.

K-means was a good method to use because it was able to handle all of the data and fit its centroid-based groupings over the principle component space. However, because of its rigid, circular boundaries, there are cutoffs that may not make as much sense when considering a continuous linguistic landscape. It also required us to choose k up front, which added an extra step and added uncertainty. We then moved on to the hierarchical clustering method. This method allows us to ease the constraints of the k-means strict circular boundaries and use a tree-like structure to find connections within and between communities. Using the dendrogram, we could visually inspect the best k to use, so that we did not have to add arbitrary human decision into the pipeline. We used the Ward method because it minimizes the variance within clusters, allowing for more tight clusters to be seen and because it is more robust to outliers. In this case, our dendrogram confirmed our k from the k-means clustering, affirming that there are generally three main dialectic groupings. However, hierarchical clustering is more complex, so we had to take only 10 percent of the responses due to its computational complexity.

When comparing the two clustering methods with the Adjusted Rand Index, we see that they have a moderately strong agreement. The adjusted rand score was 0.502, meaning that the results from the k-means and hierarchical clustering yielded clusters that mapped quite well to each other. When looking at the crosstab matrix, we can see that for the first cluster, the assignments made by each model agree over 95% of the time. This first cluster denotes the northeastern dialect, so this illustrates that both models generally agree on what locations have the northeastern dialect. There are also strong correspondences between the second and third clusters (midwest/Rust Belt and southeast regions, respectively) at 74.1% and 82.7% compatibility across these groups.

Stability of findings to perturbation

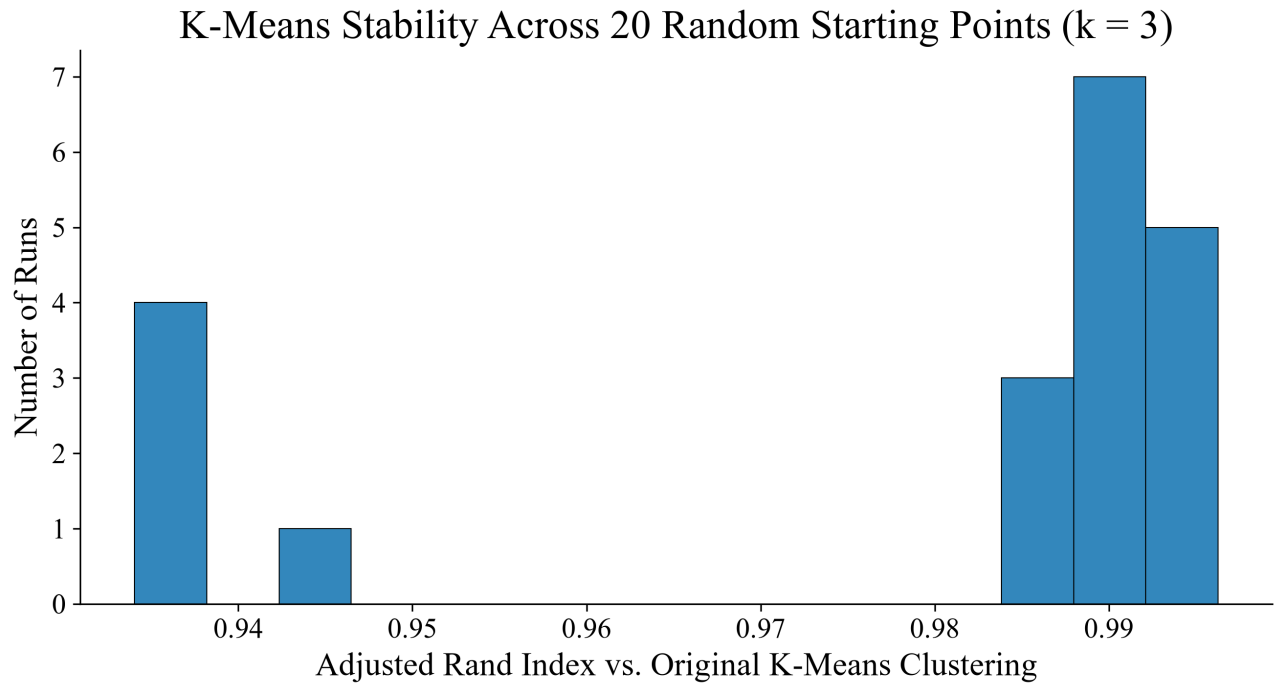


Figure 14: Histogram of K-Means Stability Runs

Table 2

	Runs	Mean ARI	Min ARI	Max ARI
K-means: different starting points	20	0.977	0.934	0.996
Hierarchical: different random samples	20	0.522	0.424	0.614

Table 6: K-Means and Hierarchical Adjusted Rand Index Scores

K-means chooses the centroids of its starting points at random, so we tested the sensitivity of different initial starting points by assigning 20 new random seeds, running the k-means model, and then using the Adjusted Rand Index to determine how similar each run was to our first mapped run performed earlier in the report. We can see that there is a little bit of a difference in some of the trials, however, the clusters found by these k-means trials were generally quite similar to our original k-means run. Then, for our hierarchical clustering, we bootstrapped the sample 20 times and then compared each bootstrapped run to the original hierarchical clustering we performed earlier. Bootstrapping essentially chooses a new set of 4600 sample points with replacement. We can see that this model still remains relatively stable, however there are greater differences between the bootstrapped runs and the original hierarchical clustering than there was difference between the k-means. This indicates a bit less stability for this model. Even if it was able to give us more refined geographic areas for our clusters, they are not necessarily as reliable as the k-means runs. In summary, the stability analysis confirmed that while hierarchical clustering provided more distinctive insights into the dialect distributions in the US, k-means serves as a more robust method, since it is more stable and reliable when perturbed.

Conclusion

The three realms of data science, the data realm, algorithms and analysis realm, and future data realm, are all relevant to the analysis done in this report. Firstly, the survey data used in this report provides a foundational understanding of how different regions have different communication and language trends across the US. It is important to note that this data was all self-reported and the users understanding of the question likely played a large roll in how they answered. Further, language can evolve over time and as people move cities, they can bring with them the dialect of their hometowns. Other factors such as place of birth, parents place of birth, social media trends, and location population density may all play a roll in these dialectical differences. The data we have is extremely helpful to create a baseline, but it can by no means give an exhaustive exploration of language differences by region. The algorithms and analysis used in this study consistently showed three large clusters of dialect: the northeast, the midwest, and the general americian baseline with distinct presence in the southeast. These algorithms can be used in conjunction with this data to add inclusivity and equity in political platforms or marketing schemes. For example, a commerical for a bubbly soda may use different language or a political message may contain different keywords based on the three main regions their campaign falls in. The algorithms seem like an effective way to test lexical trends over a large geographic area like the US. Lastly, the future data realm. Because of the stability shown by our clustering models, we can conclude that any new participant of the survey is likely to fall into one of the three main clusters found. While we can clearly see that the questions in this survey revealed lexial differences from 2006, it would be interesting to know if the trends found still apply, or if it is weaker now that there have been 20 years for humans to mix, migrate, and connect over the internet. We would hesitate to say anything concretely unless another study was performed. A reality check to verify the clustering could be to map from the hometowns of these individuals rather than their current locations. Although that data is not available, it may help control for any migration that occurs. In 2006, there were fewer online news sources, but perhaps looking at language used in paper news publications could also help verify some of the results of this analysis. The main takeaways of this analysis is that there are three distinct dialectics in the US: northeast, midwest/Rust Belt, and the general american dialect distributed throughout the south east and found largely in urban cities. Dimensionality reduction and k-means provides a robust way to visualize where dialectical differences occur throughout the US geography. Heirarchical clustering offers nice detail, but tends to have greater variance under bootstrapping. If there were more time, we would have done some manual imput for the data cleaning, especially if some geographic points were known but needed input by hand. We also would explore other clustering methods like NMF (Non-Negative Matrix Factorization), since survey results are non-negative and this method is routinely used in text mining.

Academic Honesty

Statement

Professor Bin, For these assignments, I will maintain academic integrity and cite all sources that are used including collaborations and LLM use. Academic research honesty is necessary for the ultimate trustworthiness of future scientific knowledge. Being open about assumptions made and how conclusions were drawn can ensure that generalizations are minimized and the statistics can stay as objective as possible. In a world where it can be easy to skew the statistics to support a particular point of view, research performed with integrity can help minimize the misinterpretation of the data. Further, if science was made to understand life's complexities, then it is imperative that assumptions be observed, listed and challenged. Then, we can have a more robust foundation to build future research that can withstand the test of time. We can celebrate inquiry, notice where we fail, and not let that deter us from the pursuit of something a bit greater. Being honest about research means renouncing a

fear of failure, being humble about findings, honoring the scientific method, and working toward the betterment of oneself as a scientist and the broader scientific community.

LLM Usage

LLMs were used in this lab to help with coding, much of the code was hand written, but copied over from LLM suggestion. Visualization code was copied from Claude's Opus 5.5. Certain parts of the writing were dictated out loud and Claude transcribed. Claude was told not to change the content other than to take out filler or repeated words. I also asked Claude to split, without making changes, the code and the markdown so I could have clean files for the final report. LLM Conversation links: <https://claude.ai/share/3451602f-ccf7-4edf-a5d0-e5d7eed78ad5> <https://claude.ai/share/1c069087-9466-4dbc-87c1-96e18259318c>

Collaborators

Tamar B. and I consulted about clustering methods.

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Nerbonne, J., & Kretzschmar, W. (2006). Progress in dialectometry: Toward explanation. *Literary and Linguistic Computing*, 21(4), 387–397. <https://doi.org/10.1093/lc/fql034>